

**Churn Prediction Modeling in Telecom Sector Using Python**

**Project Proposal Submitted**

**To**

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| <b>Sr. No</b> | <b>Index</b>              | <b>Page no</b> |
|---------------|---------------------------|----------------|
| 1.            | Introduction              | 3              |
| 2.            | Objectives of the Study   | 4              |
| 3.            | Scope of the Project      | 4              |
| 4.            | Work Plan and Methodology | 5              |
| 5.            | References                | 10             |

## **1. Introduction**

In today's highly competitive telecom industry, customer retention has become one of the most important challenges for service providers. Telecom companies invest significant resources in acquiring new customers, but losing existing customers can directly affect revenue and long-term business growth. Customer churn refers to the situation where customers discontinue using a company's services and shift to another service provider. Predicting customer churn has therefore become essential for telecom organizations to maintain customer loyalty and improve profitability.

With the growth of data analytics and machine learning, companies can now analyze customer behavior patterns and identify customers who are likely to leave the service. Machine learning models can help businesses take preventive actions by understanding factors responsible for customer churn. Customer information such as tenure, monthly charges, internet services, payment methods, and contract types can provide useful insights for prediction.

This project focuses on building a customer churn prediction model using Python-based machine learning techniques. The project involves data preprocessing, exploratory data analysis, hypothesis testing, feature engineering, model building, and performance optimization. Different machine learning algorithms are compared to identify the most effective model for churn prediction.

The project also uses techniques such as SMOTE for handling imbalanced data and hyperparameter tuning for improving model performance. The findings of this study can help telecom companies identify high-risk customers and improve customer retention strategies.

## **2. Objectives of the Study**

1. To study customer churn behavior in the telecom sector using customer data.
2. To perform data cleaning, preprocessing, and exploratory data analysis for understanding churn patterns.
3. To apply hypothesis testing techniques for identifying important factors affecting customer churn.
4. To build and compare different machine learning models for churn prediction.
5. To improve model performance using feature engineering, SMOTE, and hyperparameter tuning techniques.
6. To identify the most important variables responsible for customer churn prediction.

## **3. Scope of the Project**

The scope of this project is limited to customer churn prediction in the telecom sector using machine learning techniques. The project uses a telecom customer dataset containing customer demographic details, account information, services subscribed, and billing-related information.

The project focuses on analyzing customer behavior and predicting whether a customer is likely to churn or not. Various machine learning models are implemented and evaluated based on their performance metrics such as Accuracy, ROC-AUC Score, Precision, Recall, and F1-Score.

The project is developed using Python programming language and data science libraries such as Pandas, NumPy, Scikit-learn, XGBoost, LightGBM, and CatBoost.

The study can help telecom companies:

- improve customer retention,

- identify high-risk customers,
- reduce revenue loss,
- and support data-driven decision-making.

However, the project has certain limitations:

- The study is based on a single dataset.
- Real-time deployment is not included.
- External market conditions and customer psychology are not considered.
- The prediction depends only on available structured data.

## **4. Work Plan and Methodology**

The project follows a systematic machine learning workflow for customer churn prediction.

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### **Phase 1: Data Collection**

The dataset used for this project is the Telecom Customer Churn dataset. The dataset contains customer-related information such as:

- Gender
- Tenure
- Monthly Charges
- Total Charges
- Internet Services
- Contract Type

- Payment Method
- Churn Status

The dataset is imported into Python for further analysis and model development.

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## **Phase 2: Data Cleaning and Preprocessing**

Data preprocessing is performed to improve data quality and prepare the dataset for machine learning models.

The following steps are performed:

- Handling missing values
- Converting data types
- Removing unnecessary columns
- Encoding categorical variables
- Feature scaling using StandardScaler

Different encoding techniques such as binary encoding and one-hot encoding are applied based on feature types.

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## **Phase 3: Exploratory Data Analysis (EDA)**

Exploratory Data Analysis is performed to understand customer behavior and identify important patterns affecting churn.

The following analyses are carried out:

- Churn distribution analysis

- Tenure analysis
- Monthly charges analysis
- Contract type analysis
- Internet service analysis
- Correlation analysis

Graphs and visualizations are used for better understanding of the data.

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#### **Phase 4: Hypothesis Testing**

Hypothesis testing techniques are applied to determine whether certain customer attributes significantly affect churn behavior.

Statistical methods are used to test relationships between:

- Contract type and churn
- Internet services and churn
- Payment methods and churn
- Monthly charges and churn

The analysis helps identify important variables influencing customer retention.

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#### **Phase 5: Feature Engineering**

Feature engineering is performed to improve the predictive capability of machine learning models.

New features created include:

- Average Monthly Spend
- Long-Term Customer Indicator
- High Monthly Charges Indicator

These features help the models capture additional customer behavior patterns.

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### **Phase 6: Handling Imbalanced Data**

The customer churn dataset is imbalanced because the number of non-churn customers is higher than churn customers.

To address this issue, SMOTE (Synthetic Minority Oversampling Technique) is used to balance the dataset by generating synthetic samples for the minority class.

This improves the model's ability to predict churn customers effectively.

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### **Phase 7: Machine Learning Model Building**

Multiple machine learning algorithms are implemented and compared for churn prediction.

The models used in this project are:

- Logistic Regression
- Decision Tree
- Random Forest
- K-Nearest Neighbors (KNN)
- Support Vector Machine (SVM)
- Naive Bayes

- XGBoost
- LightGBM
- CatBoost

Each model is trained and evaluated using different performance metrics.

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### **Phase 8: Model Evaluation and Optimization**

The machine learning models are evaluated using:

- Accuracy Score
- ROC-AUC Score
- Precision
- Recall
- F1-Score
- Confusion Matrix

Hyperparameter tuning is performed using GridSearchCV to improve model performance.

Threshold tuning is also applied to improve recall and F1-score for churn prediction.

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### **Phase 9: Final Model Selection**

Based on model comparison and evaluation metrics, the best-performing model is selected as the final churn prediction model.

Feature importance analysis is also performed to identify the variables having the highest impact on customer churn.

## 5. References

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